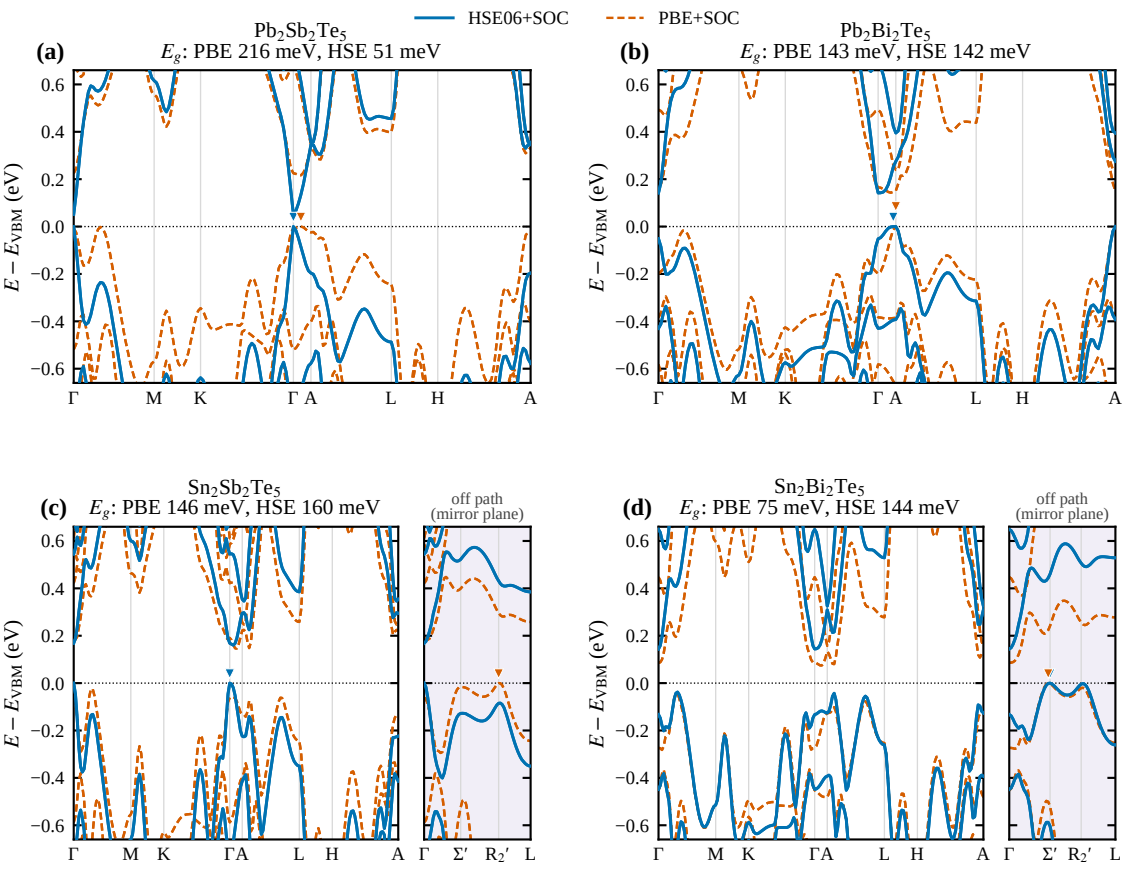

HSE06+SOC
PBE+SOC
(a)
Pb2Sb2Te5
Eg: PBE 216 meV, HSE 51 meV
(b)
Pb2Bi2Te5
Eg: PBE 143 meV, HSE 142 meV
(c)
Sn2Sb2Te5
Eg: PBE 146 meV, HSE 160 meV
(d)
Sn2Bi2Te5
Eg: PBE 75 meV, HSE 144 meV
off path
(mirror plane)
E − EVBM (eV)
0.6
0.4
0.2
0.0
−0.2
−0.4
−0.6
Γ M K Γ A L H A
Γ M K ΓA L H A
Γ Σ′ R2′ L